



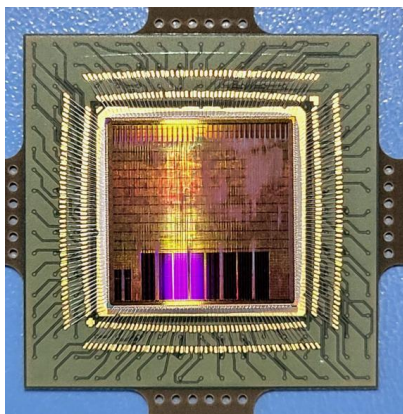
Roadmap to Printed Circuit Board Designing

You Can Build Technology That Powers Space Missions!

Imagine designing a PCB that powers a microprocessor chip destined for space! That's exactly what students at IIT Madras accomplished in collaboration with ISRO when they developed the ZVEZDA microprocessor chip.

The same PCB design skills you'll learn in this roadmap: schematic creation, component selection, layout optimization, and signal integrity, are the exact techniques used to create such revolutionary systems. What makes this even more exciting is that these students started exactly where you are now: learning the fundamentals. The PCB they designed had to handle high-speed digital signals, precise power distribution, and rigorous testing standards.

Through this three-week journey, you'll gain the foundational knowledge and practical skills to create your own amazing PCBs.



What is a PCB?

Breadboards are great for prototyping circuits, but they aren't so good for actually using the thing you're building. At some point, you'll probably want to make a project more permanent. The best way to do that is to put it on a PCB.

A Printed Circuit Board (PCB) is a flat, rigid board that mechanically supports and electrically connects electronic components using conductive tracks, pads, and other features etched from copper sheets laminated onto a non-conductive substrate. PCBs are used in virtually all electronic devices, providing the foundation for assembling complex circuits.

Types of PCBs

Type	Description
Single-Sided	One copper layer on one side of the substrate. Simple and cost-effective for basic circuits.
Double-Sided	Copper layers on both sides of the substrate, allowing for more complex routing and higher component density.
Multi-Layer	Multiple copper layers separated by insulating layers. Used for complex, high-density designs with advanced signal integrity requirements.



What is PCB Designing?

Printed Circuit Board (PCB) designing is the process of creating a layout for electronic circuits. This layout integrates all the necessary components such as resistors, capacitors, and integrated circuits, ensuring that they work together seamlessly. The design process involves several steps, including schematic capture, component placement, routing, and generating manufacturing files.

Why Learn PCB Designing?

- **Foundation of Modern Electronics:** PCBs are the backbone of virtually all modern electronic devices, from smartphones to spacecraft.
- **High-Demand Skill:** With the rise of IoT, wearable technology, electric vehicles, and smart devices, skilled PCB designers are increasingly sought after.
- **Innovation & Creativity:** PCB design enables you to create innovative electronic solutions and bring novel ideas to life.
- **Career Opportunities:** Knowledge in PCB design opens doors to roles in hardware engineering, product development, manufacturing, and R&D.
- **Project Realization:** For hobbyists and makers, PCB design is essential for transforming prototypes into professional, manufacturable products.

Prerequisites for Learning PCB Design?

Before diving into PCB design, it's beneficial to have a foundational understanding of:

1. **Basic Electronics:** Understanding components like resistors, capacitors, diodes, transistors, and their functions in a circuit.
2. **Circuit Theory:** Knowledge of how electrical circuits work, including Ohm's Law, Kirchhoff's laws, and basic circuit analysis techniques.
3. **Schematic Reading:** Ability to read and interpret electronic schematics.
4. **Computer Skills:** Familiarity with basic computer operations and software use.
5. **Mathematics:** Basic algebra and geometry skills are helpful for understanding and calculating circuit parameters.

While these prerequisites provide a helpful foundation, don't let the lack of them hold you back, what you truly need is enthusiasm and willingness to learn. Many successful PCB designers started with zero knowledge and learned everything on the job.



PCB Design Software Tools

Software	Description	Best For
KiCad	Open-source, powerful, professional-grade features	Beginners & professionals, recommended for this course
Eagle	User-friendly with extensive component libraries	Hobbyists, small projects
Altium Designer	Industry-standard, advanced features	Professional, complex designs
EasyEDA	Web-based, easy to use, integrated ordering	Quick prototypes, online collaboration

Steps to Design a PCB

Designing a PCB involves several methodical steps to ensure a functional and manufacturable electronic circuit. Here is a step-by-step guide:

1. **Define Requirements:** Determine functionality, list components, establish constraints (size, power, environmental conditions).
2. **Create Schematic:** Draw the circuit diagram, place components, verify connections and functionality.
3. **Component Placement:** Import schematic to layout tool, strategically place components considering thermal management and signal integrity.
4. **Routing:** Connect components with copper traces, optimize for signal quality and manufacturability.
5. **Design Rules Check (DRC):** Validate design against manufacturing constraints and electrical rules.
6. **Simulation & Validation:** Test circuit behavior, verify timing, power consumption, and signal integrity.

7. **Generate Manufacturing Files:** Export Gerber files, drill files, and assembly documentation.
8. **Prototype & Test:** Manufacture prototype, assemble, test functionality, identify issues.
9. **Iterate & Refine:** Make improvements based on testing, prepare for production.

By following these steps, you can systematically design a PCB that meets your project requirements and is ready for manufacturing.

Week I: Fundamentals

Focus: Master electronic components and circuit theory

□ Learning Objectives

- Understand fundamental electronic components and their behavior
- Master circuit analysis techniques and RLC circuit behavior
- Analyze frequency response, filters, and amplifier circuits

Day 1: PCB Basics & Component Introduction

Topics Covered:

- Introduction to PCB design and applications
- PCB manufacturing process overview
- Various PCB types and their use cases

Resources:

- <https://www.youtube.com/watch?v=YJr-kHy6STg&t=3>
- <https://www.youtube.com/watch?v=2Di4nCfdwgk>
- https://www.youtube.com/watch?v=GVk_hEMjzs&t=74s
- <https://www.youtube.com/watch?v=iC0LzSI2MIg>

Day 2&3: Electronic Components Deep Dive

- **Capacitors:**
Basic: <https://www.youtube.com/watch?v=X4EUwTwZ110&t=305s>

- **Inductors:** Magnetic energy storage
Basic: <https://www.youtube.com/watch?v=KSylo01n5FY>

Capacitors and Inductors Advanced:

https://www.youtube.com/watch?v=B7_yHRbCUpA

- **Transistors:**
Basic:
<https://www.youtube.com/watch?v=IcrBqCFLHIY&pp=ygUQdHJhbnNpc3RvcjBiYXNpYw%3D%3D>
https://www.youtube.com/watch?v=J4oO7PT_nzQ&t=375s&pp=ygUQdHJhbnNpc3RvcjBiYXNpYw%3D%3D

Advanced: https://www.youtube.com/watch?v=mqkVNwL_vZY

- **Diodes:**
Basic:
https://www.youtube.com/watch?v=Fwj_d3uO5g8&pp=ygULZGlvZGUgYmFzaWM%3D
Advanced: <https://www.youtube.com/watch?v=HyQOUO9QiDk>

- **MOSFETs:**
https://www.youtube.com/watch?v=AwRJsze_9m4&pp=ygUMbW9zZmV0IGJhc2lj
https://www.youtube.com/watch?v=Bfvyj88Hs_o

Day 4: Circuit Analysis

- <https://www.youtube.com/watch?v=vXQXcnRjIdU>
- <https://www.youtube.com/watch?v=ZyNYIGOaGFU>

Day 5: Bode Plots & Diode Circuits

- <https://www.youtube.com/watch?v=gNT1g0VO3yA>
- <https://www.youtube.com/watch?v=F6-EaZobHNk&pp=ygUaQm9kZSBQbG90cyBiYXNpY3MgYW5pbWF0ZWQ%3D>
- <https://www.youtube.com/watch?v=9lqwSaIDm2g&pp=ygUbRGlvZGUgQ2lyY3VpdHMgZGVzaWduIGJhc2lj>
- <https://www.youtube.com/watch?v=1uVJS5I8IC0&pp=ygUbRGlvZGUgQ2lyY3VpdHMgZGVzaWduIGJhc2lj>

Day 6: Power Supplies

- [Watch first 6 min only](#)
- <https://www.youtube.com/watch?v=RiRyzLI4Y8U&pp=ygUYYYW1wbGlmaWVyaWduIGJhc2ljIGFuaW1hdGVk>
- <https://www.youtube.com/watch?v=cK5WLMFSGnw&t=173s>

Day 7: Op-Amp Circuits

- https://www.youtube.com/watch?v=_o4ScgRZtNI&pp=ygUYYYW1wbGlmaWVyaWduIGJhc2ljIGFuaW1hdGVk
- https://www.youtube.com/watch?v=OSCKBmkrH_g

Week II: KiCad

Learning Objectives

- Install and navigate KiCad PCB design software
- Create your first schematic and PCB layout in KiCad

Day 1&2: Installation & Interface Familiarization

Quick Reference: Essential KiCad Shortcuts

Action	Shortcut
Place Component	A
Place Wire	W
Move Component	M
Rotate Component	R
Copy Component	C
Delete	Delete
Route Track	X
Start/End Track	Click
Undo	Ctrl+Z
Zoom In/Out	Mouse Wheel
Pan	Middle Mouse Button

Resources:

- <https://www.youtube.com/watch?v=rikH8tPhszk&pp=ygUNaW5zdGFsbCBraWNhZA%3D%3D>
- <https://www.youtube.com/playlist?list=PLEBQazB0HUyQ5YJSdCBb79orXaR3Uk5vm>
(watch Playlist)

Day 3&4: Schematic Basics

- https://www.youtube.com/playlist?list=PLUOaI24LpvQPls1Ru_qECJrENwzD7XImd
(watch playlist)

Day 5&6: Component Libraries & Footprints

- <https://www.youtube.com/watch?v=5NyahDfAsiE&list=PLZNH6jLLeFXsg9ohRMbJ0qqSfUrRyAn7b&index=2>
- https://www.youtube.com/watch?v=l9b_6WLemmq

follow along these!!

Day 7: Routing & Design Rules

- <https://www.youtube.com/watch?v=9HpYpP2L10A>
- <https://www.youtube.com/watch?v=9HpYpP2L10A&pp=ygUTa2ljYWQgcm91dGluZyBvdWxlcw%3D%3D>

[KiCad Official Documentation](#)

Week III: Practical PCB Design & BMS Project

Focus: Hands-on KiCad practice, component selection, and Battery Management System (BMS) design project

□ Learning Objectives

- Design complete schematics and PCB layouts in KiCad
- Understand resistor sizing, capacitor selection, and component specifications
- Learn Battery Management System (BMS) architecture and design
- Generate Gerber files for manufacturing
- Complete a real-world BMS design project

Day 1: Sound Sensor Project

Project Brief:

Design a schematic for a sound-level competition detector using Arduino Nano, sound sensor (condenser microphone), output LED, and buzzer. The system should detect sound levels and activate outputs accordingly.

Solution:

- https://drive.google.com/file/d/12Ej6OA0-zZvtjc3qcCFTlpx_riTqjmAK/view

Day 2: Tests

- <https://www.youtube.com/watch?v=inAkpoFvr0U&pp=ygUUd2hhdCBpcyBkcmMgaW4ga2ljYWTSBwkJrgoBhyohjO8%3D>
- <https://www.youtube.com/watch?v=OZhVRyuPJUA&pp=ygUUd2hhdCBpcyBkcmMgaW4ga2ljYWQ%3D>
- https://www.youtube.com/watch?v=A5uNG_jQ4qM&t=85s&pp=ygUUd2hhdCBpcyBkcmMgaW4ga2ljYWQ%3D
- <https://www.youtube.com/watch?v=Ah65M31v87c&pp=ygUUd2hhdCBpcyBkcmMgaW4ga2ljYWQ%3D>

Day 3: Simulation

- https://www.youtube.com/watch?v=ImBUSvHB_tA&pp=ygUVc2ltdWxhdGlvbiBpbiBraWQ9Y2Fk
- <https://www.youtube.com/watch?v=Gys549elAtk&pp=ygUVc2ltdWxhdGlvbiBpbiBraWQ9Y2Fk>

Day 4-7: Battery Management System (BMS) Project

Project Brief:

Modify an existing 4S BMS reference design to create a 3S-5S configurable BMS using ESP32 as the MCU. The design should include AFE for cell monitoring and balancing, fuel gauge for state-of-charge estimation, and appropriate protection circuitry.

Requirements:

- Support for 3 to 5 series-connected lithium-ion cells
- ESP32 microcontroller for processing and communication
- Cell voltage monitoring and balancing

- Overcurrent, overvoltage, and undervoltage protection
- Temperature monitoring
- State-of-charge indication

Deliverables:

- Complete schematic in KiCad
- PCB layout with proper thermal management
- Bill of Materials (BOM)
- Gerber files ready for manufacturing
- Design documentation explaining component choices

BATTERY MANAGEMENT SYSTEM(BMS) basics-

<https://www.youtube.com/watch?v=KDaBn9PLRKI>

https://www.youtube.com/watch?v=q4wDa_m9-8E

[https://www.monolithicpower.com/en/learning/resources/how-to-design-a-battery-management-system-](https://www.monolithicpower.com/en/learning/resources/how-to-design-a-battery-management-system-bms#%3A~%3Atext%3DThe%20MCU%20is%20the%20central%2Cthe%20rest%20of%20the%20system.%26text%3DThe%20AFE%20provides%20the%20MCU%2Ccurrent%20readings%20from%20the%20battery)

[bms#%3A~%3Atext%3DThe%20MCU%20is%20the%20central%2Cthe%20rest%20of%20the%20system.%26text%3DThe%20AFE%20provides%20the%20MCU%2Ccurrent%20readings%20from%20the%20battery](https://www.monolithicpower.com/en/learning/resources/how-to-design-a-battery-management-system-bms#%3A~%3Atext%3DThe%20MCU%20is%20the%20central%2Cthe%20rest%20of%20the%20system.%26text%3DThe%20AFE%20provides%20the%20MCU%2Ccurrent%20readings%20from%20the%20battery)

AFE-https://www.youtube.com/watch?v=8RV69XC_h5A&t=224s

Learn schematic and working if you are interested

Fuel Gauge-<https://www.youtube.com/watch?v=rSpxGIZQFTU>

MCU(connects AFE and fuel gauge and interfacing with rest of the system)- can be any programmable board like arduino,esp32,attiny etc

<https://simple-ee.com/2019/07/20/arduino-4s-bms-version-7/>

+ TASK- you need to modify the circuit in above link such that u use an ESP32 board as MCU and usecase of 3-5 battery.

SOLUTION

Soldering & Assembly

Good luck on your PCB design journey!

Electronics Club IITK